

Rydberg Constant Lab

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Abstract

This lab investigated the Rydberg Constant, a fundamental constant rooted within emission spectra and Bohr's atomic model. It was first determined that the Hartmann Dispersion Relation from calibration gave Hartmann Constants $b = 1.91 \pm 0.02$ and $m = 2012 \pm 6 \text{ nm}$. Using this relation an unknown gas sample was identified to be Argon. In addition, the experimental Rydberg Constant is determined to be $R_H = (1.096 \pm 0.001) \times 10^5 \text{ cm}^{-1}$, the experimental Rydberg Energy Constant is determined to be $R_{EH} = 13.59 \pm 0.02 \text{ eV}$. Finally, the yellow Sodium doublets' wavelengths are determined, giving $\lambda_1 = 588 \pm 2 \text{ nm}$, $\lambda_2 = 589 \pm 2 \text{ nm}$, with a wavelength, wavenumber and energy separation that is statistically indistinguishable from zero ($1 \pm 2 \text{ nm}$, $40 \pm 64 \text{ cm}^{-1}$, and $0.005 \pm 0.008 \text{ eV}$ respectively). However, these results may have been affected by sources of error, such as overlooking less intense spectral lines during the experiment, and taking measurements from spectral lines of varying widths.

Introduction

Before Niels Bohr

Prior to Niels Bohr's first proposal of the model of the Hydrogen Atom, the Rutherford Model was generally accepted. This planetary model described the atom having:

1. A dense, positively charged core, the nucleus, where the majority of mass resides, and
2. Light (often of negligible mass), negatively charged particles, electrons that orbit around the nucleus.

This model however had a major issue: Electrons with negative charge would radiate energy as it orbited the nucleus. Eventually, the electron shall no longer maintain the orbital action and would be pulled into the nucleus by attractive forces. [1]

Bohr's Theory of the Hydrogen Atom

In 1913, Niels Bohr first proposed his model of the Hydrogen Atom, this included the fundamental postulates:

1. Electrons reside in stationary orbits, thus they do not radiate energy.
2. Each stationary orbit has an energy level, n .
3. Photons are emitted when an electron "jumps" between energy levels, this amount is **quantized**.

In particular, Bohr quantized the **Angular Momentum** the Hydrogen Atom could have attained, given by:

$$L = n\hbar \quad (1)$$

where n represents each allowed states (energy levels), $\hbar = 1.055 \times 10^{-34} Js$ is the reduced Planck's Constant.

This leads directly to restricted orbital radii for each energy levels n , given by:

$$r = \frac{n^2 \hbar^2}{m_e k e^2} = a_B n^2 \quad (2)$$

where $m_e = 9.11 \times 10^{-31} kg$ is the electron mass, $e = 1.602 \times 10^{-19} C$ is the electron charge, $k = 8.99 \times 10^9 Nm^2/C^2$ is the Coulomb constant, $a_B = 5.29 \times 10^{-11} m$ is the Bohr Radius.

In addition, the allowed energy levels is given by:

$$E_n = -\frac{ke^2}{2a_B n^2} = -\frac{R_{EH}}{n^2} \quad (3)$$

where $R_{EH} = 13.6 eV$ is the Rydberg Energy.

In addition, Albert Einstein's discovery of the quantization of electromagnetic radiation in 1905, states that the energy of each photon is given by:

$$E = hf \quad (4)$$

where $h = 6.63 \times 10^{-34} Js$ is Planck's Constant, $f [s^{-1}]$ is the frequency of radiation. [1]

Limitations of the Bohr Model

Bohr's model proved to be a great model at the time, however it still had its limitations,

1. Although it can be extended to consider mono-electron ions such as He^+ , Li^{2+} , it fails to account for multi-electron atoms, such as He , Li .
2. The splitting of spectral lines caused by the Zeeman Effect.
3. The quantization of angular momentum is accurate for large n , but differs from observed values for smaller n .

The explanation of the phenomena mentioned above involves the use of Schrödinger's Equation and the Quantum probabilistic model of atoms.

Emission Spectra

The emission spectrum of an element is a spectrum containing discrete values of wavelength of radiation when electrons jump from a higher energy level to a lower energy level. In this experiment, different atomic gases will be ionized in a discharge tube, where it exists in a plasma state. This allows electrons traveling in the discharge tube to create vacancies in the inner or outer energy level. By considering the energy of the two states, one can expect the energy of the emitted photon to be equal to their difference.

In the Hydrogen atom, since the energy of each state is given by Equation 3, energy of each photon is given by Equation 4, this can be equated to give,

$$hf = R_{EH}(\frac{1}{m^2} - \frac{1}{n^2}) = T_m - T_n \quad (5)$$

where $m < n$ are two different energy states, T_m, T_n [eV] are the Terms (energies) of the respective state. [1]

By considering the wavelength-frequency relation of light $f = c/\lambda$, as well as fixing the state $m = 2$ Equation 5 can be rearranged to give,

$$\frac{1}{\lambda} = R_H(\frac{1}{2^2} - \frac{1}{n^2}) \quad (6)$$

where R_H [cm^{-1}] is known as the Rydberg constant, the Equation is known as the Balmer formula, discovered empirically by Johann Balmer in 1885 for the Hydrogen Atom. [1]

In addition, the Rydberg constant is then given as [1]:

$$R_H = \frac{R_{EH}}{hc} \quad (7)$$

Hartmann Dispersion

In this lab, the spectrometer is calibrated using the Hartmann Dispersion Relation. This relation relates the scale reading to the actual wavelength empirically, given by:

$$y = \frac{m}{\lambda - \lambda_0} + b \quad (8)$$

where y [unitless] is the scale reading, λ [nm] is the accepted wavelength value from literature, λ_0 [nm] is a function of the spectrometer, m [nm] and b [unitless] are fit parameters, known as the Hartmann Constants. [1]

This can also be rearranged to give wavelength as a function of experimental measurements, given by:

$$\lambda_{exp} = \frac{m}{y - b} + \lambda_0 \quad (9)$$

where λ_{exp} [nm] is the experimentally obtained wavelength.

Experimental Method

Materials

- Spectrometer with Scale and Vernier
- Sodium Lamp
- Spectral Power Supply
- Unknown Gas #16, Hydrogen and Helium Gas Tubes

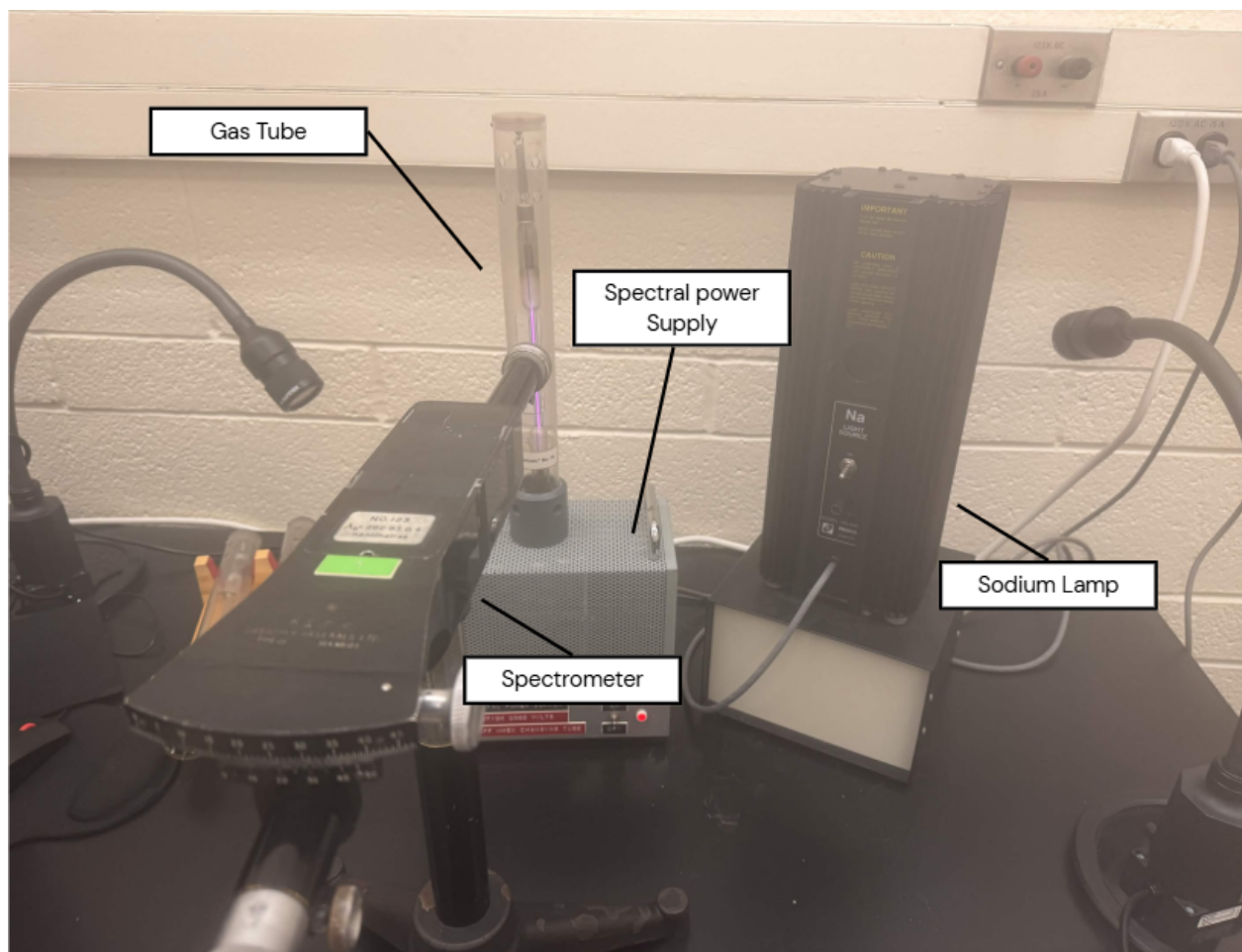


Figure 1: The setup of the lab.

Preparing the Spectrometer

1. Turn on the sodium lamp power supply and allow the colour of the light to turn yellow

2. While looking into the eyepiece, slowly rotate the eyepiece position adjustment knob until you locate bright yellow double lines.
3. Untighten the Main Focus knob, focus the spectrometer optics on the spectral line by moving the eyepiece assembly in or out.
4. Reduce the slit width until the Sodium line is clearly split into two, return to the previous step if necessary.
5. When the line is clearly focused, tighten the focusing clamp and turn off the Sodium Lamp, ensure that the paint dot on the slit adjustment is kept on the upper side of the spectrometer to maintain calibration.

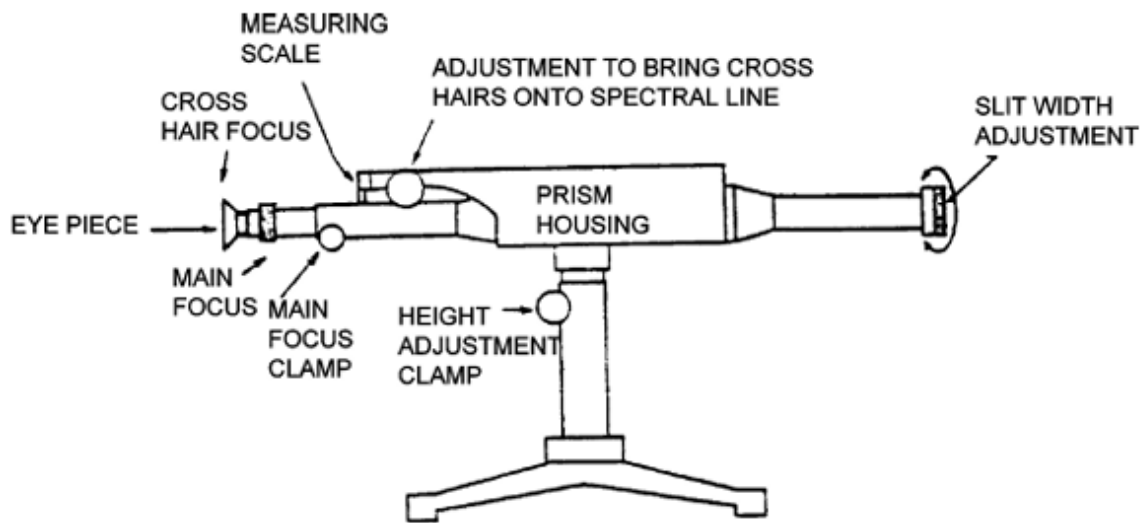


Figure 2: The structure of the spectrometer, including where the eyepiece, adjustment knobs are located at.

Taking Measurements

1. During observations, adjust the slit as desired. Opening it increases the brightness but also decreases the resolution
2. Split the line observed with the center of the cross hairs, read the scale and Vernier, then take a measurement y
3. If observations are made in the violet domain, slightly push in the eyepiece to maintain crosshair focus.

Calibration and obtaining Measurements for Hydrogen and Helium

1. Record the constant λ_0 found at the top of the spectrometer.
2. Insert the Helium gas tube and take y measurements using the described measurement procedures for each of the recommended spectral lines, found in Table 2, alongside their colours and wavelengths.
3. Repeat the process for the Hydrogen gas tube, spectral lines with colour, wavelength found in Table 3.

Obtaining measurements for Unknown Gas

1. Insert an unknown gas tube.
2. Observe and record the spectral lines seen using the described measurement procedures.

Observation of separation of Yellow Sodium doublets

1. Turn on the sodium lamp again, wait for it to warm up and turn yellow.
2. Measure and record the separation of the two yellow lines (same lines in preparing the spectrometer) using the described measurement procedures in nm , cm^{-1} and eV .

Uncertainty Propagation

From the data sheet and Lab Manual, below are the uncertainties provided [1],

$$\Delta\lambda_0 = 0.4nm \quad (10)$$

$$\Delta y = 0.01 \quad (11)$$

In addition, since provided standard wavelengths in the lab manual aside from those for sodium are accurate to $0.1nm$, we estimate the uncertainty of the known wavelengths to be:

$$\Delta\lambda = 0.1nm \quad (12)$$

Noting that λ are known theoretical wavelengths, not the propagated one, and that this uncertainty doesn't apply to theoretical sodium wavelengths. [1]

For propagating uncertainties in calculations in general, the following formula is used:

$$\Delta z = \sqrt{\sum_{i=1}^N \left(\frac{\partial f}{\partial x_i} \Delta x_i \right)^2} \quad (13)$$

where Δz is the uncertainty of the result, f is the function used to calculate the result, N is the number of parameters in the function, x_i is the i th parameter of f , and Δx_i is the

uncertainty associated with the i th parameter. [2]

Thus, in accordance with Eqns. 9-13, the calculated wavelengths from scale readings has propagated uncertainty given by:

$$\Delta\lambda_{exp} = \sqrt{\left(\frac{m}{y-b}\right)^2 \left[\frac{\Delta y^2 + \Delta b^2}{(y-b)^2} + \left(\frac{\Delta m}{m}\right)^2 \right] + \Delta\lambda_0^2} \quad (14)$$

In addition, the uncertainty of constant variable product can be expressed as follows:

$$\Delta y = c\Delta x \quad (15)$$

Where x and y are variables with uncertainties denoted with Δ , c is a constant, and $y = cx$. [2]

The uncertainty of a reciprocal of a variable can also be expressed as follows:

$$\Delta y = \frac{\Delta x}{x^2} \quad (16)$$

Where x and y are variables with uncertainties denoted with Δ , and $y = \frac{1}{x}$. [2]

Finally, the uncertainty propagation formula for addition and subtraction is:

$$\Delta c = \sqrt{(\Delta a)^2 + (\Delta b)^2} \quad (17)$$

where Δ denotes the uncertainty of the associated variables such that $c = a \pm b$. [2]

Raw Experimental Data

$$\frac{\lambda_0 \text{ [nm]}}{282.8 \pm 0.4}$$

Table 1: Hartmann Dispersion function of the spectrometer, λ_0 found on top of the spectrometer.

Note that the violet band with the smaller wavelength in the hydrogen spectrum was not successfully observed, this will be discussed in the Error Section.

Results and Analysis

Part 1, Hartmann Relation Calibration

Since uncertainties are present for both x and y values, Orthogonal Distance Regression (ODR) is used in Python to fit data from Tables 2-3 into Equation 8. Fig 3 is obtained.

λ [$\pm 0.1nm$]	Colour	y [± 0.01 unitless]
447.1	Blue	14.16
471.3	Blue	12.61
492.2	Blue	11.50
501.6	Cyan	11.13
587.6	Yellow	8.50
667.8	Red	7.15

Table 2: Accepted theoretical helium wavelengths, λ ; colours extracted from the provided table; Scale values, y obtained from measurements. The uncertainties for λ and y are in accordance with Eqns. 11 12.

λ [$\pm 0.1nm$]	Colour	y [± 0.01 unitless]
410.2	Violet	Unknown
434.0	Violet	15.15
486.1	Blue	11.82
656.3	Red	7.28

Table 3: Accepted theoretical hydrogen wavelengths for the Balmer series, λ ; colours extracted from the provided table; Scale values, y obtained from measurements. The uncertainties for λ and y are in accordance with Eqns. 11 12. In addition, one of the violet spectral lines could not be found due to its low intensity, so its scale reading y is unknown.

Colour	y [± 0.01 unitless]
Purple	17.00
Purple	16.55
Purple	15.50
Purple	15.20
Purple	13.85
Purple	13.30
Purple	13.05
Purple	12.63
Blue	10.93
Cyan	10.50
Cyan	10.45
Green	9.44
Green	9.16
Yellow	8.43
Orange	8.15
Red	6.70
Red	5.81

Table 4: Observed colours and corresponding scale reading y values for Unknown Gas #16. The uncertainty for y is in accordance with Eqn. 11.

Colour	y [± 0.01 unitless]
Yellow	8.47
Yellow	8.50

Table 5: Scale values y for observed yellow sodium spectral lines. The uncertainty for y is in accordance with Eqn. 11.

This fit is quite accurate as R^2, χ_v^2 are both approximately equal to 1, indicating a suitable fit. In addition, residuals are scattered with both positive and negative values, showing a random pattern, not demonstrating any trend arising from errors. Thus the Hartmann Constants are carried over in Parts 3 and 4, given by Eqns. 18 19.

$$b = 1.91 \pm 0.02 \quad (18)$$

$$m = 2012 \pm 6nm \quad (19)$$

Part 2, Identification of Unknown Gas

To obtain the unknown gas, 2 factors are considered, including its emitted colour when heated, as well as the spectral lines matching with known data.

Firstly, as shown in Figure 4, the colour of the Unknown Gas is purple when a voltage source is applied to the gas tube. This matches the colours of argon and nitrogen when heated in a low-pressure environment. This means the unknown gas is either argon or nitrogen. [1]

In addition, from Table 6, it is observed that all except one of the λ_{exp} values have a corresponding matching argon wavelength from the accepted argon spectrum within two uncertainties. In contrast, most of the λ_{exp} values don't have a corresponding nitrogen wavelength from the accepted nitrogen spectrum within two uncertainties.

Thus, we conclude that the Unknown Gas #16 in this case is most likely Argon.

Part 3, Rydberg Constant

To find the Rydberg constant, the scale values y of hydrogen from Table 3 were converted to their experimental wavelength values using Eqns. 9 14 and the Hartmann constants 18 19. The energy states n were then determined for the experimental wavelengths, and the data was plotted in accordance with Eqn. 6.

From Fig. 5, it is determined that $R_H = (1.096 \pm 0.001) \times 10^7 m^{-1} = (1.096 \pm 0.001) \times 10^5 cm^{-1}$. Using Eqns. 7 15, the experimental Rydberg energy constant and its uncertainty can be determined as $R_{EH} = (2.178 \pm 0.002) \times 10^{-18} J = 13.59 \pm 0.02 eV$. This result agrees with the theoretical Rydberg energy constant of around 13.6 eV [1], as the theoretical value is within one uncertainty of the experimental value.

λ_{exp} [nm]	Argon Wavelength Match [$\pm 0.1nm$]	Nitrogen Wavelength Match [$\pm 0.1nm$]
416.1 ± 0.6	415.9	416.4
420.2 ± 0.6	420.1	420.1
430.8 ± 0.7	430.0	—
434.2 ± 0.7	433.5	434.4
451.3 ± 0.8	451.1	—
459.4 ± 0.8	459.6	—
463.4 ± 0.8	462.8	464.9
470.5 ± 0.8	470.2	472.4
505.8 ± 1.0	—	—
517.0 ± 1.1	516.2	—
518.4 ± 1.1	518.8	—
550.0 ± 1.3	549.6	—
560.3 ± 1.4	560.7	—

Table 6: Experimental wavelengths λ_{exp} with closest matching argon and nitrogen spectral lines (within $2\Delta\lambda_{exp}$ of the experimental value). The uncertainties for argon and nitrogen wavelengths are in accordance with Eqn. 12. Experimental wavelengths were calculated from the scale readings y of Table 4 using Eqns. 9–14 with Hartmann constants 18–19. Only the experimental wavelengths for green, cyan, blue and violet colours are included for comparison with known spectra (Tables 10–11).

λ_{exp} (nm)	$\Delta\lambda_{exp}$ (nm)	$1/\lambda_{exp}$ (m^{-1})	$\Delta(1/\lambda_{exp})$ (m^{-1})	n	$\frac{1}{2^2} - \frac{1}{n^2}$
434.8	0.7	2.300×10^6	4×10^3	5	0.2100
485.8	0.9	2.058×10^6	4×10^3	4	0.1875
657	2	1.521×10^6	5×10^3	3	0.1389

Table 7: This table contains the experimental values of hydrogen wavelengths of the Balmer Series and its uncertainties, as well as the reciprocal of the experimental wavelengths with uncertainties propagated using Eqn. 16. The corresponding energy states n for the wavelengths are also listed, as well as values for $\frac{1}{2^2} - \frac{1}{n^2}$. By Eqn. 6, this is the data needed to determine R_H .

The physical significance of R_{EH} is that it is the ionization energy of an electron in the ground state of hydrogen [3]. Since the experimental R_{EH} was determined to agree with the theoretical value, the experimental R_{EH} is also close to the ionization energy of atomic hydrogen. In addition, R_{EH} also determines the spacing between the quantized energy levels of hydrogen according to the Bohr model by Eqn. 5, and by extension, the emission spectra of hydrogen.

Part 4, Separation of Sodium doublets

From Table 5, Eqns. 9–14 were used with Hartmann constants 18–19 to determine the experimental wavelengths λ_{exp} for the yellow sodium spectral lines. The results are summarized in Table 8.

y [unitless]	λ_{exp} [nm]	λ [± 0.0001 nm]
8.47 ± 0.01	588 ± 2	588.9950
8.50 ± 0.01	589 ± 2	589.5924

Table 8: The experimental wavelengths for the yellow sodium spectral lines from Table 5, the corresponding λ_{exp} propagated from scale values y , and the matching accepted theoretical λ values. Note that uncertainties for theoretical sodium wavelengths are assumed to be 0.0001 nm, as the sodium wavelengths are accurate to 0.0001 nm. [1]

From Table 8, it is observed that the two accepted theoretical wavelength values λ for the spectral lines of sodium are within one uncertainty of their corresponding experimental values λ_{exp} . Thus, it can be concluded that both the calibration performed in Part 1 and the measurements taken from the spectrometer had a high degree of accuracy.

Furthermore, from Table 8, the separation of the two yellow lines can be expressed in nm , cm^{-1} and eV . These results are summarized in Table 9.

Quantity	Value	Uncertainty
$\Delta\lambda$ [nm]	1	2
$\Delta\tilde{\nu}$ [cm^{-1}]	40	64
ΔE [eV]	0.005	0.008

Table 9: Measured wavelength separation and corresponding values in wavenumber (spatial frequency) $\Delta\tilde{\nu}$ and energy ΔE , where $\Delta\tilde{\nu} = \Delta\frac{1}{\lambda}$ [4], and Δ denotes differences in quantities between the two sodium spectral lines. These quantities were calculated with experimental wavelengths from Table 8 using Eqn. 4 and the wavelength-frequency relation $f = c/\lambda$ where c is the speed of light in a vacuum. Uncertainties were also propagated using Eqns. 15 16 17 applied to the relationships between wavelength, wavenumber, energy, and the differences in the respective quantities between the two sodium spectral lines.

From Table 9, it is noticed that the uncertainties of the observed wavelength, wavenumber and energy separations of the yellow spectral lines of sodium are greater than their respective values. This means that the observed separation between the sodium spectral lines is statistically indistinguishable from zero, and more refined focusing or a spectrometer with a higher resolving power would be needed to produce a statistically significant result for the separation of the yellow sodium spectral lines.

Error Sources

Aside from random errors introduced from measurements, as noted by the Lab Manual, different spectral lines have varying intensities [1]. If a spectral line has a low relative intensity, it is potentially missed and could affect data collection. In our case, a purple spectral line is missed for Hydrogen which reduces the degree of freedom by 1, causing the Rydberg constant to only be dependent on 3 collected data points. In addition, the Argon/Nitrogen emission spectrum consists of a large amount of spectral lines. When measuring the spectral lines of the Unknown Gas, it is unavoidable that dimmer ones are missed, affecting the confidence of

drawing the conclusion that the Unknown Gas is Argon.

In addition, in order to view dimmer lines, the slit size can be adjusted. As noted, a slit more opened leads to clearer but "thicker" lines, which affects the accuracy of determining the position where the crosshair "splits" the spectral line into half, altering measurement results.

Conclusion

Overall, this lab determined an experimental Rydberg Constant as well as related wavelength measurements explained by Bohr's atomic model. First, a Hartmann Dispersion Relation from calibration was done to produce Hartmann Constants $b = 1.91 \pm 0.02$ and $m = 2012 \pm 6$ nm. Using this relation an unknown gas sample's emission spectrum was analyzed and determined to be Argon. In addition, the experimental Rydberg Constant is determined to be $R_H = (1.096 \pm 0.001) \times 10^5 \text{ cm}^{-1}$, the experimental Rydberg Energy Constant is determined to be $R_{EH} = 13.59 \pm 0.02 \text{ eV}$, both closely resembling the theoretical value. Finally, the yellow sodium doublets' wavelengths are determined, giving $\lambda_1 = 588 \pm 2 \text{ nm}$, $\lambda_2 = 589 \pm 2 \text{ nm}$, which is similar to the literature values provided. However, the separation between the sodium doublet is statistically zero, with a wavelength separation of 1 ± 2 nm, wavenumber separation of $40 \pm 64 \text{ cm}^{-1}$, and energy separation of $0.005 \pm 0.008 \text{ eV}$. All these results may have been affected by sources of error, such as overlooking less intense spectral lines during the experiment, as well as taking measurements from spectral lines of varying widths depending on the adjusted spectrometer slit sizes.

Appendix

λ [$\pm 0.1nm$]	Colour
574.0	Green
565.0	Green
560.7	Green
557.3	Green
549.6	Green
522.1	Green
518.8	Green
516.2	Green
470.2	Blue
462.8	Blue
459.6	Blue
452.2	Purple
451.1	Purple
433.5	Purple
433.4	Purple
430.0	Purple
426.6	Purple
425.9	Purple
420.1	Purple
419.8	Purple
416.4	Purple
415.9	Purple

Table 10: Argon Wavelength, λ and colours in range 415.9 to 574.0 nm. The uncertainty for argon wavelengths is in accordance with Eqn. 12. [1]

λ [$\pm 0.1nm$]	Colour
497.6	Green
491.7	Green
481.5	Green
472.4	Blue
466.7	Blue
464.9	Blue
457.4	Blue
449.0	Blue
441.7	Violet
435.5	Violet
434.4	Violet
427.0	Violet
420.1	Violet
414.2	Violet
409.5	Violet
406.0	Violet
399.8	Violet
394.3	Violet

Table 11: Nitrogen Wavelength, λ and colours in range 394.3 to 497.6 nm. The uncertainty for nitrogen wavelengths is in accordance with Eqn. 12. [1]

References

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- [3] D. Sadoway, “Energy, Frequency, Wavelength, and Ionization,” Chemistry LibreTexts, MIT OpenCourseWare. [Online]. Available: https://chem.libretexts.org/Bookshelves/Inorganic_Chemistry/Introduction_to_Solid_State_Chemistry/06%3A_Recitations/6.04%3A_Energy_Frequency_Wavelength_and_Ionization
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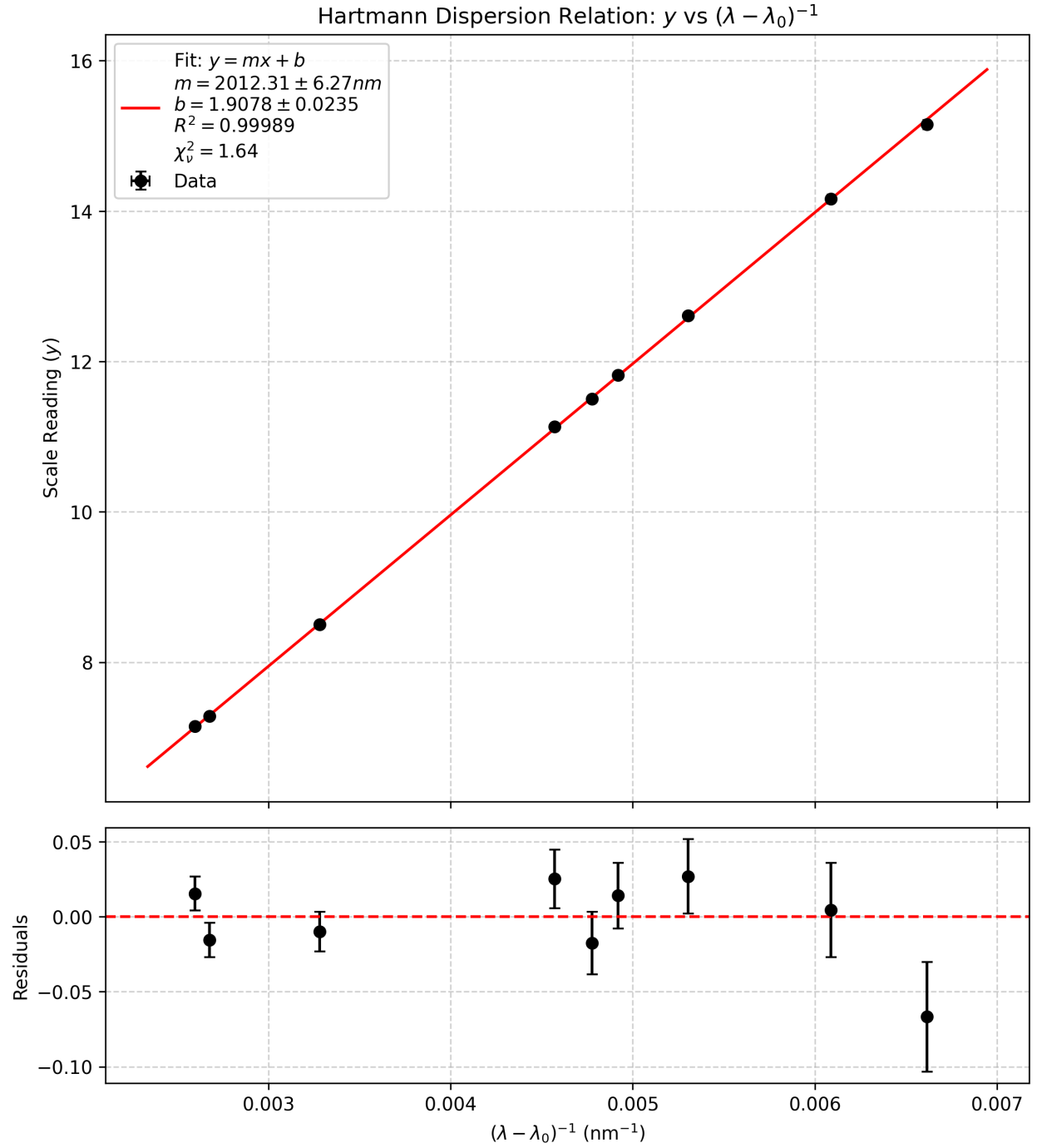


Figure 3: Graph and residuals for scale readings y [unitless] vs $(\lambda - \lambda_0)^{-1} [\text{nm}^{-1}]$, using data from Tables 2 3 and fit with an Orthogonal Distance Regression (ODR). Error bars are present, but some are too small to be seen. The slope, $m = 2012 \pm 6 \text{ nm}$; intercept, $b = 1.91 \pm 0.02$ [unitless]. $R^2 = 0.99989$, which indicates that the linear regression is an almost perfect fit. $\chi^2_v = 1.64 > 1$, which implies that the uncertainties of each data point were slightly underestimated (see the Error Sources section for additional sources of error).

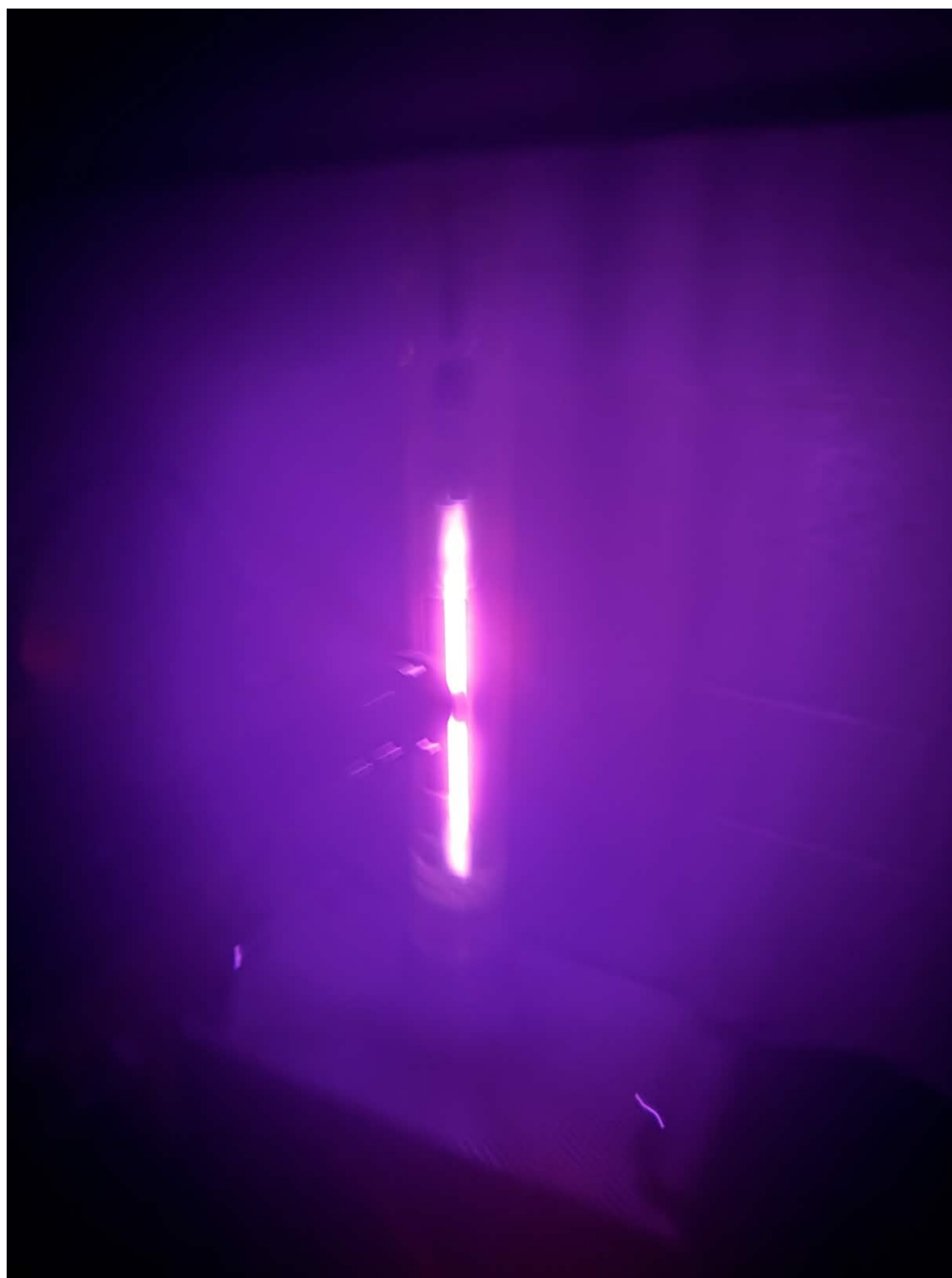


Figure 4: The Unknown Gas' purple colour when heated.

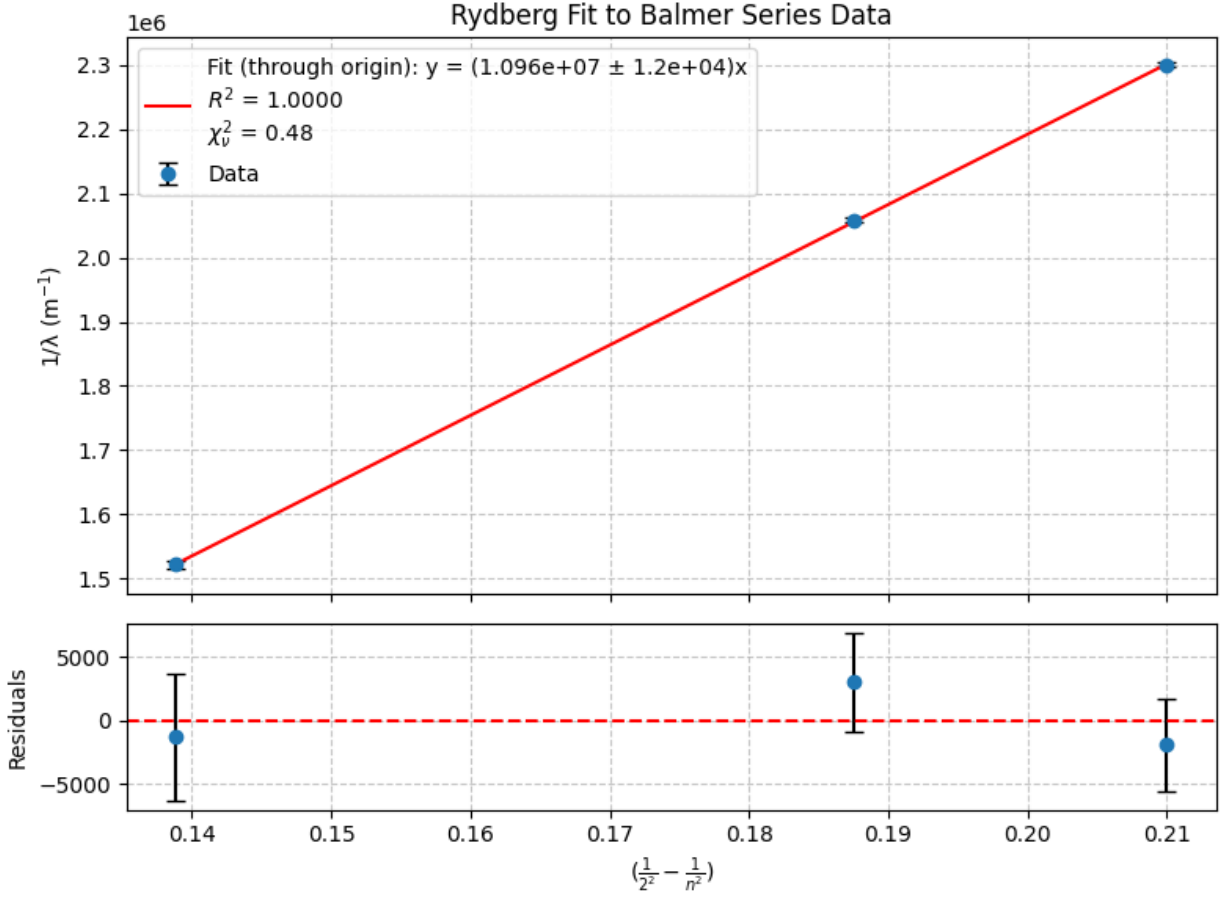


Figure 5: Graph and residuals for $1/\lambda_{exp}$ vs $\frac{1}{2^2} - \frac{1}{n^2}$, using data from Table 7 and fit with a least squares linear regression through the origin in accordance with Eqn. 6. Error bars are present, but some are too small to be seen. The slope R_H is $(1.096 \pm 0.001) \times 10^7 \text{ m}^{-1}$. In addition, $R^2 = 1.0000$, which indicates that the linear regression is a perfect fit; and $\chi^2_\nu = 0.48 < 1$, which implies that the uncertainties of each data point were overestimated. However, these goodness of fit metrics may not be very reliable given the small sample size of 3, and the low χ^2_ν value may be a result of overfitting to the small sample size (see the Error Sources section for the explanation for the small sample size).